

A Model for Magnetization Transfer in Tissues

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Magnetization transfer in several tissues is measured and successfully modeled using a two-pool model of exchange. The line shape for the semi-solid pool is characterized by a superLorentzian and the liquid pool by a Lorentzian. The tissues investigated were white and gray matter, optic nerve, muscle, and liver. All tissues the authors studied are characterized by the same model but differ in the parameter values of the model. Blood and cerebral spinal fluid (CSF) were also investigated. The two-pool model with a Lorentzian line shape for both the semi-solid and liquid pools modeled the magnetization transfer in blood. In CSF, as expected, there is no measurable exchange of magnetization. The T_{2B} associated with the semi-solid pool was short ($\sim 10 \mu s$) for all tissues indicating a fairly rigid semi-solid pool. In addition characterization of the line shape as superLorentzian indicates molecules such as integral membrane proteins or lipids in membranes are likely molecules participating in the exchange. Conversely, in blood large globular proteins are indicated due to the Lorentzian nature of the semi-solid pool and a $T_{2B} \approx 300 \mu s$.

Key words: magnetization transfer; tissues; superLorentzian; model.

INTRODUCTION

Spin exchange between water protons and the semi-solid component of tissue (i.e., magnetization transfer) has provided a relatively new imaging technique (1). Some clinical applications of magnetization transfer are angiography (2), white matter disease (3–5), metastatic lesions (6), the musculoskeletal system (7), cartilage (8), and improvement in contrast of brain lesions while using paramagnetic contrast agents (9).

Despite the useful clinical applications of magnetization transfer, magnetization transfer in tissues is still not well understood at a fundamental level. There have been several attempts to characterize the behavior of the mobile and semi-solid protons in a variety of systems, but so far no comprehensive picture has emerged. In this paper we present a two-pool magnetization transfer model for several different systems (white and gray matter, muscle, liver, optic nerve, blood, and cerebral spinal fluid (CSF)) and extract the parameters from the experimental data that characterize the exchange and relaxation times.

The first two-pool model for magnetization transfer was the Bloch equations with exchange given by McConnell (10). There are two pools: a liquid and a semi-solid pool with Lorentzian NMR line shapes characterizing each pool. This was the first model used to extract information on the rate of exchange between the two pools for systems such as agar (11), lipid bilayers (12), rat muscle (13), calf lens homogenate (14), heat denatured egg albumin (15), and cross-linked bovine serum albumin (16). However, in all the above systems only one off-resonance irradiation power was used and, as was eventually realized, this is not sufficient for uniquely determining the model parameters or drawing any conclusions about the validity of the model (17).

It was noted by Swanson (18) that the implicit assumption in the Bloch equations that the semi-solid pool had a Lorentzian line shape was not valid for immobile spins, which are coupled to one another through the homonuclear dipole-dipole Hamiltonian. Rather, a Gaussian line shape characterizes spins that are homogeneously coupled and this line shape was subsequently used to successfully model magnetization transfer for various concentrations of agar (17). Several irradiation powers were evaluated and are required in order to uniquely determine the model parameters and rigorously test the model. Recently a two-pool model showing a transition from Lorentzian to Gaussian for the semi-solid line shape with an increase in bovine serum albumin concentration has been published (19).

Another model for magnetization transfer has been proposed involving three pools of spins: liquid, moderately mobile, and motionally restricted (20). This was used to model magnetization exchange in calf lens nuclear homogenate and cross-linked bovine serum albumin and, once again, several offset frequencies and powers were used. This model introduces five additional parameters, over that required for the two-pool model.

Our approach to analyzing magnetization transfer in tissue was to try to keep the model as simple as possible and to limit the number of free parameters. We decided to use the two-pool model and look for an appropriate semi-solid line shape. As demonstrated in Fig. 1, neither of the two standard line shapes, i.e., Lorentzian and Gaussian, adequately characterizes the semi-solid component of the two-pool model for bovine white matter. (Details of the experiment and fit are given in subsequent sections: the solid lines are best fits of the model for the two line shapes.) However, we did find a line shape for the semi-solid pool that adequately describes all the tissues we studied—the superLorentzian lineshape (21)—and we present the results in this paper. We also show, for the different tissues we have studied, that they are all characterized by the same model, differing only in adjustment of the various rates of the model.

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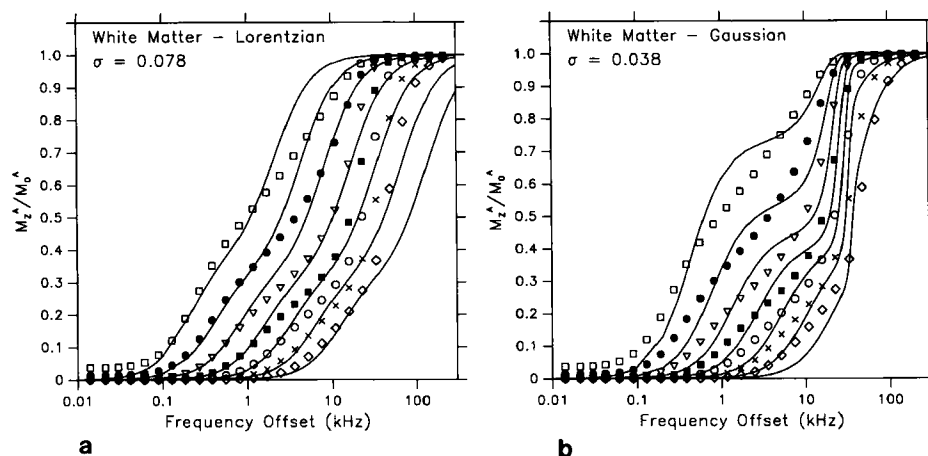


Fig. 1. Magnetization transfer experiments for bovine white matter. M_z^A/M_0^A is the ratio spins in pool A aligned longitudinally to the total number of A spins. Data are shown for seven different RF amplitudes ($\omega_1/2\pi = 83$ ($\square$), 170 ($\bullet$), 330 ($\triangle$), 670 ($\blacksquare$), 1340 ($\circ$), 2670 ($\times$), and 5340 ($\diamond$) Hz) and 27 different offset frequencies. In (a) solid lines are a fit of Eq. [3] with a Lorentzian line shape for the semi-solid pool, while in (b) a Gaussian line shape is used. Parameters are given in Table 1. The average residual deviation per point is 0.078 for the Lorentzian and 0.038 for the Gaussian.

THEORY

Model

The model used to analyze all our magnetization transfer experiments is shown in Fig. 2, which is identical to that presented in ref. 17. The system is modeled as having two pools: a liquid (pool A) and a semi-solid, i.e., macromolecular, (pool B). Each pool has an independent longitudinal relaxation rate, R_A and R_B , where $R = 1/T_1$. In the presence of off-resonance irradiation, some longitudinal magnetization in each pool is lost (i.e., the direct

effect). R_{rfA} and R_{rfB} are the rates associated with this loss and depend on the amplitude ω_1 ($\omega_1 = \gamma H_1$, where H_1 is the applied field strength) and offset frequency Δ of the irradiation pulse:

$$R_{rfi} = \omega_1^2 \pi g_i(2\pi\Delta) \quad [1]$$

where $g_i(2\pi\Delta)$ is the absorption line shape for the spins in pool i (see line shape section below), Δ is in Hz and ω_1 is in rads/sec (18). The rate of transfer of magnetization between the two pools is characterized by a fundamental rate constant R . The rate of exchange in either direction depends on the concentrations of both pool A (M_0^A) and pool B (M_0^B), which remain constant in size. Therefore, the rate of exchange in either direction is a second order rate constant given by $R M_0^B M_0^A$. By convention, the number of spins in pool A, M_0^A , is normalized to 1, thus the rate of exchange reduces to $R M_0^B$, but this does not imply a first order rate constant as has been suggested by others. Formulated this way, we have a rate constant which is a fundamental parameter of the exchange probability and is independent of the concentration of the semi-solid compartment (17).

The liquid pool is generally assumed to have a Lorentzian line shape $g(2\pi\Delta)$ characterized by a width $1/T_{2A}$ satisfying $(2\pi\Delta T_{2A})^2 \gg 1$ for all significant non-zero points. Under these assumptions, the direct effect for the A spins is

$$R_{rfA} = \left(\frac{\omega_1}{2\pi\Delta} \right)^2 \frac{1}{T_{2A}}. \quad [2]$$

For a detailed discussion of this model we refer the reader to refs. 13, 16, and 17, where the differential equations and solutions describing these processes can be found. In this paper, for conciseness, we only give the

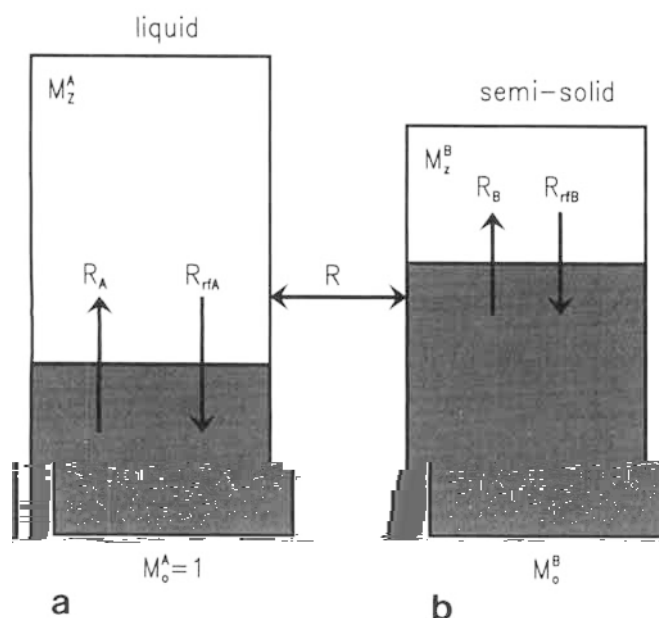


Fig. 2. The two-pool model for magnetization transfer. The shaded region in each pool represents spins that are not aligned longitudinally. All rates and populations are defined in the text.

solution for the longitudinal component of the A spins, M_{z^A} :

$$M_{z^A} = \frac{R_B \left[\frac{RM_0^B}{R_A} \right] + R_{rfB} + R_B + R}{\left[\frac{RM_0^B}{R_A} \right] (R_B + R_{rfB}) + \left(1 + \left(\frac{\omega_1}{2\pi\Delta} \right)^2 \left[\frac{1}{R_A T_{2A}} \right] \right) (R_B + R_{rfB} + R)} \quad [3]$$

There are five unique parameters to be determined: R , R_B , $[RM_0^B/R_A]$, $[1/R_A T_{2A}]$, and T_{2B} (via R_{rfB}). As pointed out by Caines *et al.* (13), we cannot obtain unique model parameters R_A , T_{2A} , R_B , T_{2B} , R , and M_0^B because experiments are performed in the steady state and thus do not contain explicit reference to absolute time scales. In the case where there is no magnetization transfer (i.e., $R = 0$, $RM_0^B = 0$), Eq. [3] reduces to the direct effect, i.e., the effect an off-resonance irradiation pulse has on the liquid pool alone.

Line Shapes

In tissues, the motions of the spins largely determine the NMR properties. For protons, motions manifest themselves through the modulation of the spin-spin dipolar interactions. In a rigid system these interactions, which are static, result in a line shape that is approximately Gaussian. However, in the presence of motions, which average these interactions out, the resonance line is narrowed and has a Lorentzian shape. The expressions for these two line shapes are

$$\text{Lorentzian: } g(2\pi\Delta) = \frac{T_2}{\pi} \frac{1}{[1 + (2\pi\Delta T_2)^2]} \quad [4]$$

$$\text{Gaussian: } g(2\pi\Delta) = \frac{T_2}{\sqrt{2\pi}} e^{-\frac{(2\pi\Delta T_2)^2}{2}}. \quad [5]$$

Liquids are a typical example of motional narrowing and thus pool A is assumed to be Lorentzian. To date, published work has described the semi-solid pool as either Lorentzian or Gaussian. However, for tissues neither of these line shapes give acceptable fits for the magnetization transfer data (Fig. 1). A line shape that does work well in tissues is the superLorentzian.

The superLorentzian is the line shape that is expected to arise from partially ordered materials, for example, polymers, liquid crystals, biological membranes or molecules absorbed on surfaces (21). The superLorentzian line shape was first observed in potassium laurate (21) and subsequently in lamellar liquid crystals (22) and lipids (23). It has also been seen as part of the complex proton NMR spectrum of whole cells (24). To understand the origin of this line shape, we need to look at the interactions that influence the NMR spectrum more closely.

The proton spin Hamiltonian is primarily the sum of Zeeman, $\mathcal{H}_z$, chemical shift, $\mathcal{H}_{cs}$, and dipolar, $\mathcal{H}_d$. The proton NMR spectrum can be interpreted in terms of the

approximate Hamiltonian

$$\mathcal{H} = \mathcal{H}_z + \mathcal{H}_{cs} + \overline{\mathcal{H}_d} + h(t) \quad [6]$$

where $\overline{\mathcal{H}_d}$ is the time-averaged dipolar Hamiltonian and $h(t) = \mathcal{H}_d(t) - \overline{\mathcal{H}_d}$, the term that gives rise to broadening of the spectral lines. The broadening may be taken as Lorentzian in character if the correlation time for $h(t)$ is short. If the molecule is undergoing rapid, isotropic reorientation, (i.e., $M_2\tau_c^2 \ll 1$, where M_2 is the second moment associated with $\mathcal{H}_d$), then $\overline{\mathcal{H}_d} = 0$ and the proton spectrum is governed by chemical shift and broadening effects due to $h(t)$. An example of such molecules are proteins or peptides in solution (24). However, integral membrane proteins and lipid molecules in membranes normally have a non-zero time-averaged dipolar Hamiltonian $\overline{\mathcal{H}_d}$ (24). In the case of lipids, the nonisotropic, planar symmetry of the membrane ensures that $\overline{\mathcal{H}_d} \neq 0$, despite the appreciable motional averaging that takes place. The rapid rotation of the phospholipid molecule in a membrane bilayer about its long axis results in a projection of the dipolar interactions along the symmetry axis of the motion, i.e., the normal to the bilayer. The observed superLorentzian spectrum has been explained in terms of the scaling of the dipolar Hamiltonian with the angle θ between the external magnetic field and the bilayer normal (22, 23),

$$\overline{\mathcal{H}_{d,\theta}} = \overline{\mathcal{H}_{d,\theta=0}} \left(\frac{3\cos^2\theta - 1}{2} \right). \quad [7]$$

This gives the following line shape when all θ are present

$$g(2\pi\Delta) = \int_0^1 \frac{3}{|3\cos^2\theta - 1|} f\left(\frac{4(2\pi\Delta)}{|3\cos^2\theta - 1|}\right) d(\cos\theta) \quad [8]$$

where f can be any function that characterizes the line shape at $\theta = 0$, such that $\int_{-\infty}^{\infty} f(x)dx = 1$. If f is Gaussian, as frequently assumed (22–24), Eq. [8] becomes

$$g(2\pi\Delta) = \int_0^{\pi/2} d\theta \sin\theta \sqrt{\frac{2}{\pi}} \frac{T_2}{(3\cos^2\theta - 1)} e^{-2\left[\frac{2\pi\Delta T_2}{3\cos^2\theta - 1}\right]^2}. \quad [9]$$

EXPERIMENTAL CONSIDERATIONS

NMR Measurements and Analysis

For all our studies we used a spectrometer (SMIS, Surrey, England) with a 30-cm, horizontal bore superconducting magnet (Nalorac, Martinez, CA) operating at 1.5 T. Our continuous wave magnetization transfer pulse sequence was composed of an irradiation pulse of amplitude H_1 at offset frequency Δ applied for 7 s followed by a 90° pulse after a delay time of 200 μ s. The 90° pulse duration was 10 μ s. The irradiation pulse was 7 s long to ensure steady state had been reached (17) and for cerebral spinal fluid was extended to 13 s due to the long T_1 (2.3 s). Data acquisition started 20 μ s after the 90° pulse and ten 12-bit data points were acquired at a rate of 10 μ s per point and averaged to improve the signal-to-noise ratio. The time between data acquisition and the start of the next satu-

ration pulse was set to 5 s. Phase cycling the 90° pulse and data acquisition were done to compensate for any transverse magnetization following the off-resonance irradiation.

To test magnetization transfer models and uniquely determine fitting parameters, the magnetization of the liquid pool must be measured at several different powers and offset frequencies of the off-resonance irradiation pulse. We used 27 different offset frequencies Δ , distributed on a logarithmic scale from 0.014 to 213 kHz. The experiment was performed at seven different irradiation powers ω_1 , increasing from 0.083 to 5.34 kHz by factors of 2. The amplitudes of the irradiation pulses were calibrated by determining the length of a rectangular pulse required to invert the spin population.

Our magnetization transfer figures (for example Fig. 1) are plots of M_z^A/M_0^A versus Δ , on a logarithmic scale. The powers and corresponding symbols are listed in the caption of Fig. 1. Reproducibility of normalized magnetization transfer signals was better than 0.5%.

The magnetization transfer data is fit to Eq. [3] with $R_{\tau/B}(2\pi\Delta) = \pi\omega_1^2 g(2\pi\Delta)$, where $g(2\pi\Delta)$ is the semi-solid line shape: either Lorentzian (Eq. [4]), Gaussian (Eq. [5]) or superLorentzian (Eq. [9]). Each line shape is characterized by one parameter T_{2B} . For the superLorentzian line shape the integral is evaluated numerically. The five parameters; R , R_B , $[RM_0^B/R_A]$, $[1/R_A T_{2A}]$ and T_{2B} , are determined using the downhill simplex method, a sum of squares minimization routine (25). The sum of squares is

$$ssq = \sum_{i=1}^N [\text{fit}(x_i) - \text{data}(x_i)]^2 \quad [10]$$

where N is the number of data points. Quality of the fit is evaluated as the average residual deviation per point, σ , between the fit and experiment, ($\sigma = \sqrt{ssq/N}$). The error of a fitted parameter is determined by finding the maximum and minimum value of the parameter for an increase of 1 in χ^2 , while keeping all other parameters fixed.

SAMPLES

The white and gray matter, optic nerve, muscle, and liver samples were bovine, and the blood and cerebral spinal fluid (CSF) samples were human. The white matter was taken from the corpus callosum. The experiments were all done within 5 h postmortem and the samples were maintained at 4°C before the experiments. The blood and CSF experiments were performed within 30 min of extraction. The samples were sealed in test tubes with parafilm. Each set of experiments took approximately 1.5 h. Before and after the magnetization transfer experiment, the multicomponent T_2 was measured using a CPMG sequence to ensure samples had not degraded or changed appreciably during the course of the experiment. No changes in the T_2 spectrum were observed. All experiments were done at room temperature (20–22°C). In the case of blood, the sample was gently vortexed between each different RF power experiment, (approx-

mately every 12 min), to reduce separation of plasma and red blood cells.

RESULTS

Figure 3 shows the fit of bovine white matter magnetization transfer data to the two-pool model with a super-Lorentzian line shape for the semi-solid pool. The fitted parameters are given in Table 1 along with the fitted parameters for the Lorentzian and Gaussian line shapes. There is a large decrease in the average residual deviation per point, σ , for the superLorentzian. We would like to emphasize that we did not use a χ^2 statistic because we are interested in the validity of the structure of the model and do not believe that the residual deviations between the measurements and model arise from experimental error. A further improvement in σ will be unlikely to alter the fitted parameters significantly, as evidenced in Table 1, where the parameters R and RM_0^B/R_A do not vary much with line shape. The T_{2B} for the Lorentzian line shape is more than an order of magnitude larger than for the Gaussian: a large T_2 is necessary for the Lorentzian to be narrow.

There are two regions of data to which we draw your attention. The first is the 83 Hz amplitude at low Δ (<100 Hz). The values in that region are higher than we expect due to B_0 inhomogeneities over the sample, and these five points were not included in the fit. The second region is at the other end of the spectrum, large Δ and an amplitude of 5.34 kHz. These points were included in the fit, but our model could not successfully account for the attenuated signal intensity at these points. A minor adjustment in the line shape at frequencies >70 kHz could probably account for this. This would have minimal impact on the fitted model parameters or model validity and thus we decided to use well-characterized and phys-

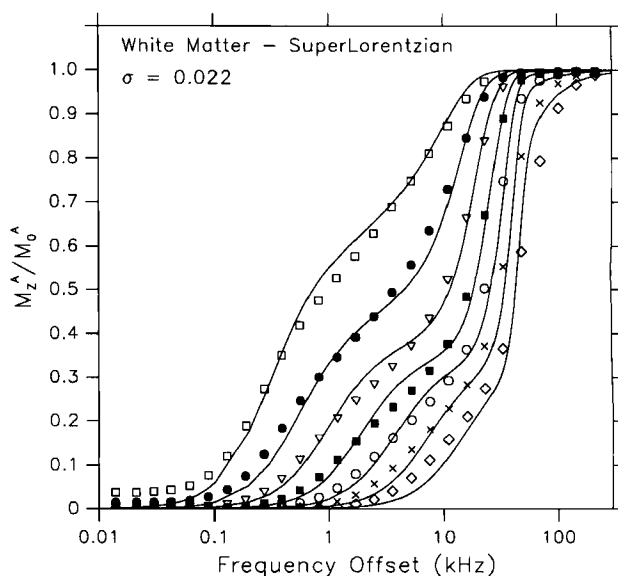


Fig. 3. The same data as in Fig. 1 fitted to Eq. [3] with a super-Lorentzian line shape for the semi-solid pool. Parameters are given in Table 1. The average residual deviation per point is 0.022.

Table 1
Parameters for the Fit of Eq. [3] to the White Matter Magnetization Transfer Data for Three Different Semi-Solid Line shapes. $R_B = 1 \text{ s}^{-1}$

Line shape (White Matter)	$R \text{ (s}^{-1}\text{)}$	$\frac{RM_o^B}{R_A}$	$\frac{1}{R_A T_{2A}}$	$T_{2B} \text{ (}\mu\text{s)}$	σ
Lorentzian	20 ± 2	2.1 ± 0.1	17 ± 3	230 ± 20	0.078
Gaussian	16 ± 3	1.5 ± 0.1	40 ± 3	18.0 ± 0.7	0.038
SuperLorentzian	21 ± 3	2.0 ± 0.1	23 ± 4	10.4 ± 0.5	0.022

ically significant line shapes such as the superLorentzian.

We tested the superLorentzian line shape on the agar data of ref. 17 which used a Gaussian line shape to characterize the semi-solid pool. We found the superLorentzian and Gaussian gave similar average residual deviation per point for the 2% concentration, but the superLorentzian was somewhat worse for the 4 and 8% (i.e., 0.020 compared with 0.016).

The data and fits for several more tissues are given in Fig. 4, (optic nerve, gray matter, liver, and muscle). The two-pool model with the superLorentzian line shape for the semi-solid pool is quite good for a distribution of tissue types. For comparison, σ values for fits to the two-pool model using Lorentzian, Gaussian and superLorentzian line shapes shown in Table 2. The parameters associated with the superLorentzian fit are given in Table 3. They are ordered according to the maximum magnetization transfer, (Max MT). Max MT is defined as the maximum difference between the direct effect M_Z^A/M_o^A

Table 2
Average Residual Deviations, σ , for Best Fits of Eq. [3] Using Lorentzian, Gaussian, and superLorentzian Semi-Solid line shapes

Tissue	Average Residual Deviation σ		
	Lorentzian	Gaussian	SuperLorentzian
Muscle	0.068	0.030	0.018
White matter	0.078	0.038	0.022
Optic nerve	0.076	0.035	0.021
Gray matter	0.067	0.034	0.019
Liver	0.042	0.035	0.021
Blood	0.009	0.041	0.035
CSF	0.010	0.010	0.010

(with $R = 0$), (i.e., if there was no exchange after the off-resonance irradiation) and M_Z^A/M_o^A with magnetization transfer. Table 3 lists Max MT, which is determined using the experimental data points, and also lists the model Max MT, which uses the fitted model parameters.

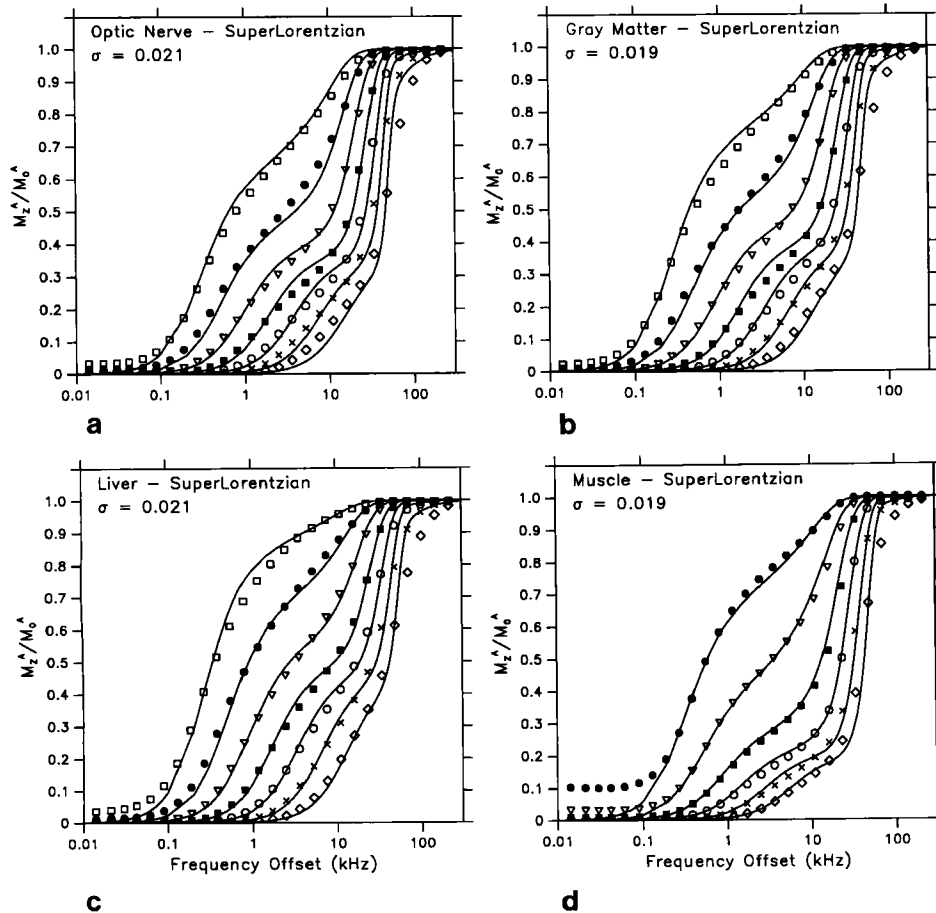


Fig. 4. Magnetization transfer data for optic nerve (a); gray matter (b); liver (c); and muscle (d). The solid lines are fit to Eq. [3] with a superLorentzian semi-solid line shape. Fit parameters are given in Table 3.

Table 3
Parameters for the Fit of Eq. [3] to the Magnetization Transfer Data for Each of the Tissues Investigated

Tissue	Exper. Max MT	Model Values			Fitted Parameters			
		Max MT	$\Delta_{\max}$ (kHz)	$2\pi\omega_{1\max}$ (Hz)	R (s ⁻¹)	$\frac{RM_0^B}{R_A}$	$\frac{1}{R_A T_{2A}}$	T_{2B} (μs)
Muscle	0.66	0.67	12.3	1500	300 ± 30	4.3 ± 0.3	5.4 ± 0.8	7.6 ± 0.3
White matter	0.54	0.55	10.0	550	21 ± 3	2.0 ± 0.1	23 ± 4	10.4 ± 0.5
Optic nerve	0.53	0.55	11.1	610	20 ± 3	1.9 ± 0.1	22 ± 3	9.5 ± 0.5
Gray matter	0.53	0.52	11.3	750	32 ± 4	1.7 ± 0.1	16 ± 2	9.2 ± 0.4
Liver	0.43	0.44	13.3	1020	53 ± 6	1.3 ± 0.1	13.4 ± 1.4	7.8 ± 0.5

Semi-solid line shapes are superLorentzian. Max MT is calculated using the data and $[1/R_A T_{2A}]$ from the model for the direct effect. Model Max MT is calculated using the model parameters to find the optimum Δ and power for the irradiation pulse. $R_B = 1 \text{ s}^{-1}$.

The offset frequency and power at which the maximum occurs varies with tissue and is also listed in the table for the model Max MT. The Δ is relatively constant at slightly more than 10 kHz, but the maximum power varies by a factor of 3 over the range of tissues and is significantly different from the value of 5 μT ($2\pi\omega_1 \approx 200 \text{ Hz}$), which has been recommended for continuous wave (CW) magnetization transfer experiments.

As expected RM_0^B/R_A follows the same trend as Max MT: what is important for the magnitude of magnetization transfer is the exchange rate RM_0^B in relation to the longitudinal relaxation rate, R_A . $1/R_A T_{2A}$ is the direct effect contribution and the fit is most sensitive to this term when $R_{\text{eff}B}$ is large and pool B is completely saturated so that Eq. [3] then depends only on the direct effect. The fitted value for $[1/R_A T_{2A}]$ is much different from the value obtained by measuring average values of R_A and T_{2A} using inversion recovery and CPMG sequences, respectively, in the absence of an irradiation pulse. Therefore, there is no way to uniquely determine R_A and T_{2A} . Some expressions have been published relating the R_A measured independently and that obtained from the fit (26) but do not take into account the multi-component relaxation behavior (27) and in our case no additional insight into the model was gained by attempting to separate R_A and T_{2A} . T_{2B} for all tissues is quite short ($\approx 10 \mu\text{s}$). This value is similar to Swanson's 9 μs T_2 for denatured egg white (18), and Henkelman *et al.*'s value of 12.9 μs for agar (17) where Gaussian line shapes were assumed. Bloom *et al.*'s second moment determina-

tion using a superLorentzian line shape (24) give a T_2 of 8 μs for the integral membrane protein rhodopsin and T_2 s of 9 μs (gel phase) and 27 μs (fluid phase) for phospholipids.

The results for blood, which are shown in Fig. 5, are quite different from all other tissues presented. In this case a superLorentzian is not the best semi-solid line shape ($\sigma = 0.03$). Rather a Lorentzian is the best line shape ($\sigma = 0.009$). This is not surprising given the much more fluid composition of blood as compared with tissue. A possible semi-solid pool in this system would be the large globular proteins such as albumin or hemoglobin. According to Bloom *et al.* (24) large globular proteins in water undergo rapid, isotropic reorientation and $\mathcal{H}_d = 0$, resulting in a Lorentzian line shape. Bloom *et al.* estimate that such proteins would have T_2 values on the order of 2 ms. This indicates the semi-solid component in blood, characterized by a Lorentzian with at T_2 of 0.28 ms, would be more restricted than large free proteins in water, consistent with the presence of more complex structures such as blood cells.

Figure 5b shows the graph of magnetization transfer data and fit for CSF. In this system there is essentially no magnetization transfer. We fit the curves using the Lorentzian line shape, and the model does indeed show R and RM_0^B/R_A are immeasurably small (see Table 4). The solid lines in the figure are essentially identical to the direct effect curves determined by independent measurements of T_1 ($=2.26 \text{ s}$) and T_2 ($=2.25 \text{ s}$) of the CSF.

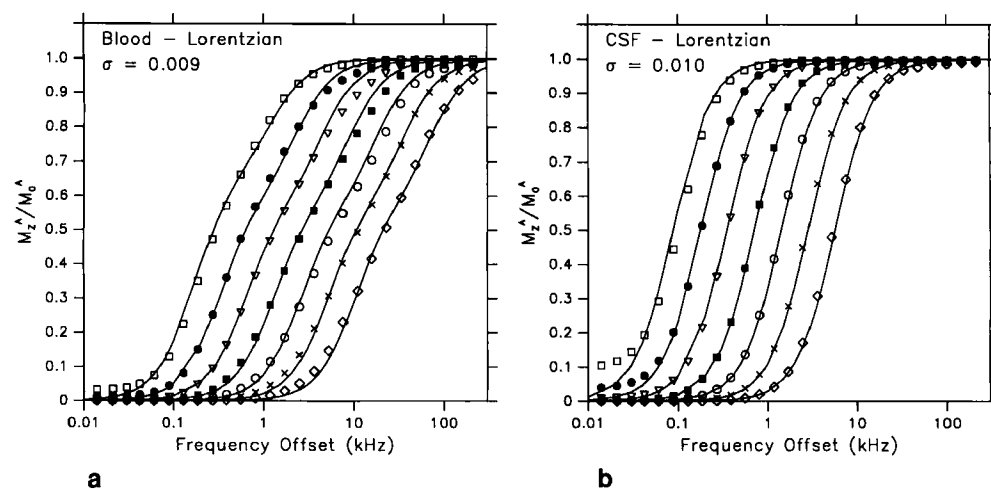


Fig. 5. Magnetization transfer data for blood (a) and cerebral spinal fluid (b). The solid lines are fits to Eq. [3] with a Lorentzian line shape. The residual deviation per point is 0.009 for blood and 0.010 for CSF. All parameters are given in Table 4.

Table 4
Parameters for the Fit of Eq. [3] to the Magnetization Transfer Data for Blood and Cerebral Spinal Fluid (CSF)

System	Semi-solid Line shape	Max MT	R (s^{-1})	$\frac{RM_0^B}{R_A}$	$\frac{1}{R_A T_{2A}}$	T_{2B} (μs)
Blood	Lorentzian	0.26	32 ± 4	0.70 ± 0.05	6.1 ± 0.6	280 ± 40
CSF	Lorentzian	<0.01	$<1 \times 10^{-3}$	$<1 \times 10^{-4}$	1.12 ± 0.06	

There is no measurable magnetization transfer for CSF and M_z^A can essentially be accounted for by direct effect only. For blood, Max MT is essentially obtained for all powers but at varying Δ . $R_B = 1 s^{-1}$.

Pike *et al.* (28) have reported ratios for the amount of signal remaining after pulsed saturation to the amount in the absence of the pulse, for blood, muscle, and gray and white matter on a 1.5 T whole-body imaging system. (Due to hardware constraints of most clinical imagers, the semi-solid proton pool is saturated using short, intense on-resonance RF pulses (29).) Our Max MT value reflects the signal suppression with magnetization transfer taking place with respect to the signal suppression due to the off-resonance irradiation pulse effect on the liquid spins. As shown in Fig. 6, Max MT compared with 1 minus the MT ratio reported by Pike *et al.* agree very well. Hajnal *et al.* (30) have reported clinical MT values (ratio of signal with saturation pulse to without pulse) for white and gray matter and muscle at 0.15 T using an off-resonance irradiation pulse ($\Delta\nu = 5$ kHz, $B_1 = 17 \mu T$, i.e. $2\pi\omega_1 = 680$ Hz) designed to produce substantial change in the tissue signal while avoiding undue direct saturation of the liquid pool. Figure 6 shows that 1 minus Hajnal *et al.* MT ratios compare well with Max MT.

DISCUSSION

The two-pool model for magnetization transfer in the tissues works well when the semi-solid pool line shape is characterized by a superLorentzian. The superLorentzian line shape has been observed in biological systems such

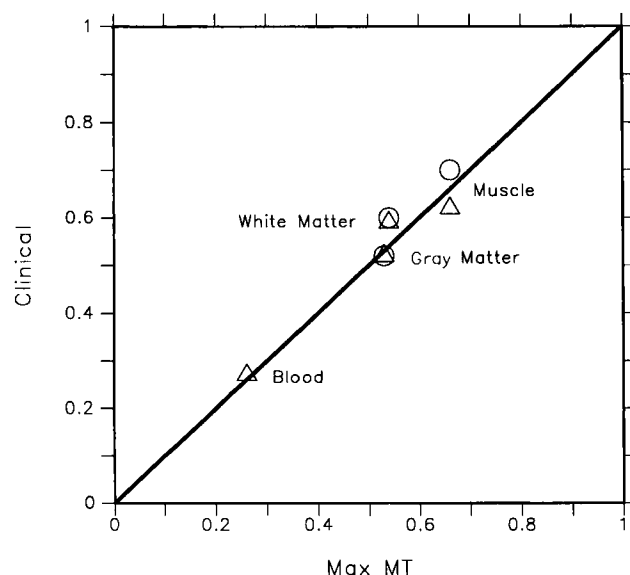


Fig. 6. Comparison of our Max MT measurement and one minus the ratio of signal remaining after a binomial pulse to the amount in the absence of the pulse obtained by Pike *et al.* (28) Δ and one minus the ratio of signal with saturation pulse to without pulse obtained by Hajnal *et al.* (30) $\circ$ on clinical imagers.

as phospholipids, and can be understood from the non-zero averaging of the dipolar Hamiltonian, thus giving us confidence in its validity. During the course of our investigation we tried several formulations for the line shape, such as Lorentzian and Gaussian convolutions, but none worked as well as the superLorentzian. It is possible there is a line shape that could give a smaller average residual deviation per point, but we would not expect much change in the two-pool model exchange and relaxation rates because the two-pool model with a superLorentzian line shape tracks the magnetization transfer data well, with only slight deviations from the data (unlike the Lorentzian and Gaussian line shapes). The total semi-solid line shape could be directly measured in such systems; however, the measured line shape would include all protons in the semi-solid not just the ones that are participating in magnetization transfer, thus limiting its usefulness for interpreting magnetization transfer experiments.

The blood and CSF give quite different magnetization transfer results. For blood the two-pool model also works well but the semi-solid line shape is characterized as Lorentzian as opposed to superLorentzian. Thus, the semi-solid pool in blood might be composed of large globular proteins. CSF has virtually no magnetization transfer and hence the line shape is not determined.

Recently there have been suggestions in the literature that the dipolar longitudinal relaxation rate, R_{1d} , for the semi-solid component is important during continuous irradiation (31, 32). We investigated this and found an improvement in the fit at powers of 330, 670, and 1340 Hz and offset frequencies between 2 and 10 kHz, but the average residual deviation did not improve significantly (for white matter σ went from 0.022 to 0.019, and for the other tissues, the improvement with the addition of the dipolar reservoir was even less). The exchange and relaxation rates did not change significantly with the addition of the parameter R_{1d} (33).

Through looking at a variety of tissues, we have searched for fundamental magnetization transfer differences in different tissues. For liver, muscle, optic nerve, and white and gray matter, the same model for magnetization transfer performs equally well in all systems. There are quantitative differences in the rates of exchange and RM_0^B relative to R_A , which are consistent with the maximum amount of magnetization transfer we observe. Our T_{2B} values are all quite short, which implies a fairly rigid and similar structure for the semi-solid pool, perhaps integral membrane proteins or lipid headgroups. Therefore, clinical use of magnetization transfer effects for tissue specific contrast will need to exploit quantitative differences in rate constants for differing tissues.

Hopefully, the quantitative model presented here will provide a foundation for such studies.

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